KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION COMMUNICATION TECHNOLOGY

TEST

CYS 121 INFORMATION SYSTEM SECURITY

Duration 3 Hours

INSTRUCTIONS

1. **Answer 5 Questions**
2. **Any reference material is not permitted**

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Question One

Briefly explain the following Cyber security terminologies and discuss their differences. Vulnerability, threat, Attack and Integrity [5 Marks each]

Question Two

At times attack may happen without the organization realizing the extent of the damage. You are required to explain with examples the circumstance in which a real damage and high cost to the Information System(IS) may be regarded to have occurred. Explain along these lines of reasoning

a) Criticality of information attacked [5 Marks]
b) Availability of the IS [5 Marks]
c) Possibility of recovery [5 Marks]
d) Cost to the organization [5 Marks]

Question Three

Explain at least three kinds of harm that could be experienced from electronic unauthorized access and modification of confidential materials in the following scenarios:

a) In a competitive business world [5 Marks]
b) Examining body [5 Marks]
c) Health Institution [5 Marks]
d) Courts of law [5 Marks]

Question Four

Attempting to break into (that is, obtain access to or use of) a computing system without authorization should be illegal. Argue for or against this statement with reasons in terms of benefits(if you argue for) or damages (if you argue against) that might be caused. [20 Marks]

Question Five

Briefly discuss five(5) Categories of threats with examples. [4 Marks Each]

Question Six

Consider a program that allows consumers to order products from the web or online. Who might want to attack the program? What types of harm might they want to cause? What kinds of vulnerabilities might they exploit to cause harm? [20 Marks]

Question Seven

There are several ways to authenticate data subjects (users) of a system,

a) You are required to identify (5) of them and explain what they are [10 Marks]
b) Giving examples show advantages and disadvantages of each [10 Marks]

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